

# Sebastian Viegas

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## EDUCATION

### University of Notre Dame

*Bachelor of Science in Aerospace Engineering | Minor: Engineering Corporate Practice*  
GPA: 3.64/4.00

Notre Dame, IN  
Expected May 2028

University of Notre Dame Global Gateway Study Abroad

Rome, Italy  
May - June 2025

## EXPERIENCE

### Mach 10 Free-Piston Reflected Shock Tunnel Undergraduate Researcher

January 2026 - Present

*Corke Lab*

*Notre Dame, IN*

- Modeled free-piston shock dynamics and tunnel operational boundaries using Queensland L1d4 and NASA CEA to predict thermochemical expansion through the nozzle at static conditions ranging from 5-10 MPa and 1850-3000 K.
- Performed high-fidelity 2D axisymmetric CFD simulations with over 2 million cells as small as 1e-6 m in ANSYS Fluent to analyze exit conditions and boundary layer behavior across nozzle contours.
- Ran FEA simulations to ensure adequate factor of safety of 4 under extreme dynamic loads for both static and dynamic parts.
- Finalized facility CAD assembly in SolidWorks within 600 sqft laboratory constraints; prepared high-pressure components for DFMA, supplier sourcing, and manufacturing quotes.

### Mechanisms Lead & Subsystem Placement Responsible Engineer

September 2024 - Present

*IrishSat*

*Notre Dame, IN*

- Leading a student team in end-to-end design, analysis, and fabrication of moving satellite hardware, including reaction wheels, deployable solar panels, and monopole antennas.
- Architected IrishSat's first fully modular CubeSat design and full-system SolidWorks assembly, modeling subsystem components to validate physical fit and system tolerances.
- Designed reaction wheels engineered for satellite modularity; completed torque and dynamic rotational calculations to verify the satellite could rotate on any axis within 7 deg/s.
- Executed MATLAB thermal simulations of orbital solar radiation on battery housings; constructed a scale 2U physical prototype via laser cutting and 3D printing to validate internal component packaging.

### UCAH Hypersonic Projectile Flight Competition Project Team Lead

September 2025 - Present

*Corke Lab*

*Notre Dame, IN*

- Appointed Team Lead for an upcoming hypersonic flight competition after co-authoring a design report on a high L/D (3+), low-altitude waverider projectile design.
- Developed MATLAB scripts to parametrically iterate projectile geometries and analyzed CFD performance in ANSYS Fluent to ensure geometries resulted in a minimum L/D of 3 at various angles of attack.
- Fabricated 3D-printed models to experimentally validate shock angles and surface thermal loads in a Mach 6 Ludwig wind tunnel.

### Experimental Liquid Propulsion Team Member

September 2025 - May 2026

*Notre Dame Rocketry Team*

*Notre Dame, IN*

- Designed a custom graphite nozzle insert for liquid propulsion Engine Genesis to withstand 3350 K and 400 psi pressure loads within the throat and combustion chamber.
- Built a static test stand for the hybrid Engine Alpha, machining assembly components via waterjet cutting and horizontal bandsaw.
- Performed dynamic FEA simulations in Fusion 360 under 200 lbf thrust load conditions to optimize structural geometry and ensure safety margins.

## SKILLS

**Software:** ANSYS Fluent, MATLAB, SolidWorks, Fusion 360, NASA CEA, Queensland L1d4, Python, Java, Microsoft Suite

**Fabrication & Machining:** 3D Printing, Waterjet Cutting, Laser Cutting, Machine Shop Tools, GD&T

**Aerospace Domains:** Hypersonic Aerodynamics, CFD & FEA Simulation, Orbital Mechanisms, Gas Dynamics, Satellite Integration, Liquid Propulsion

**Languages:** Advanced Portuguese, Intermediate French

**Interests:** Piano, Hurdling, Music Production